DO NOW — RECALL

Lesson 154 · Search Engines: Crawling, Indexing and Re

Name: _____ Class: _____ Date: _____

1. Why does a server recheck a browser form?

2. When can binary search be used?

3. What data structure follows LIFO?

4. What is a database foreign key?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Client checks are bypassable.

2. Model answer

On sorted data.

3. Model answer

Stack.

4. Model answer

A reference to a key in another table.

By the end, you can…

• Crawler fetches pages found through links and sitemaps.

• Map each searchable term to candidate pages.

• Match query terms and rank relevant results.

Vocabulary — know these terms

crawler — Search agent fetching linked pages.

inverted index — Terms mapped to document identifiers.

tokenisation — Breaking text into searchable units.

ranking — Ordering candidate search results.

NEW LEARNING

Read, annotate and explain

Discover

Crawler fetches pages found through links and sitemaps.

Index

Map each searchable term to candidate pages.

Retrieve

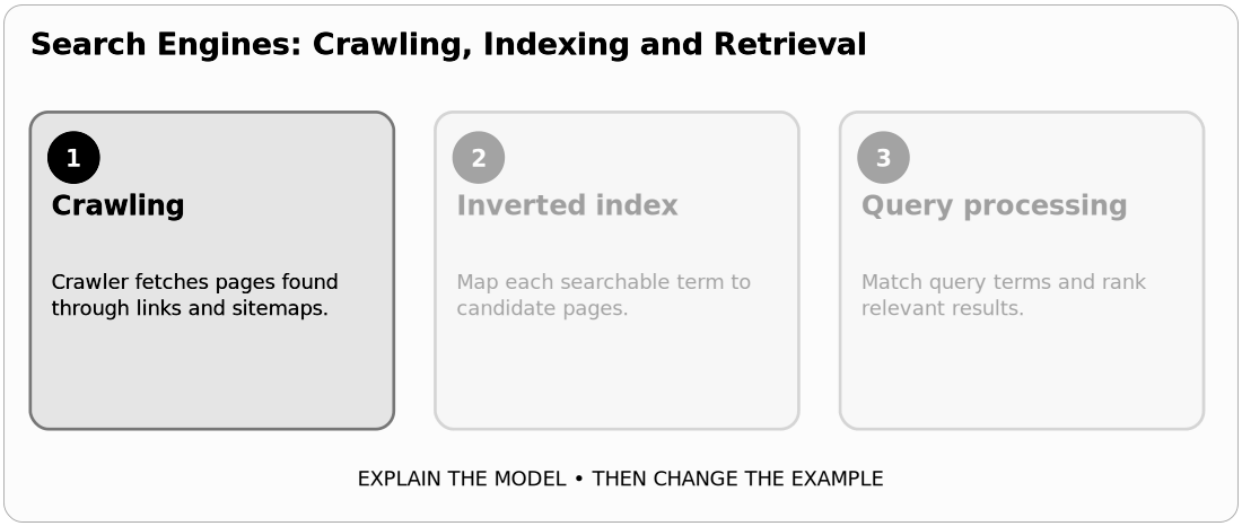
Match query terms and rank relevant results.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

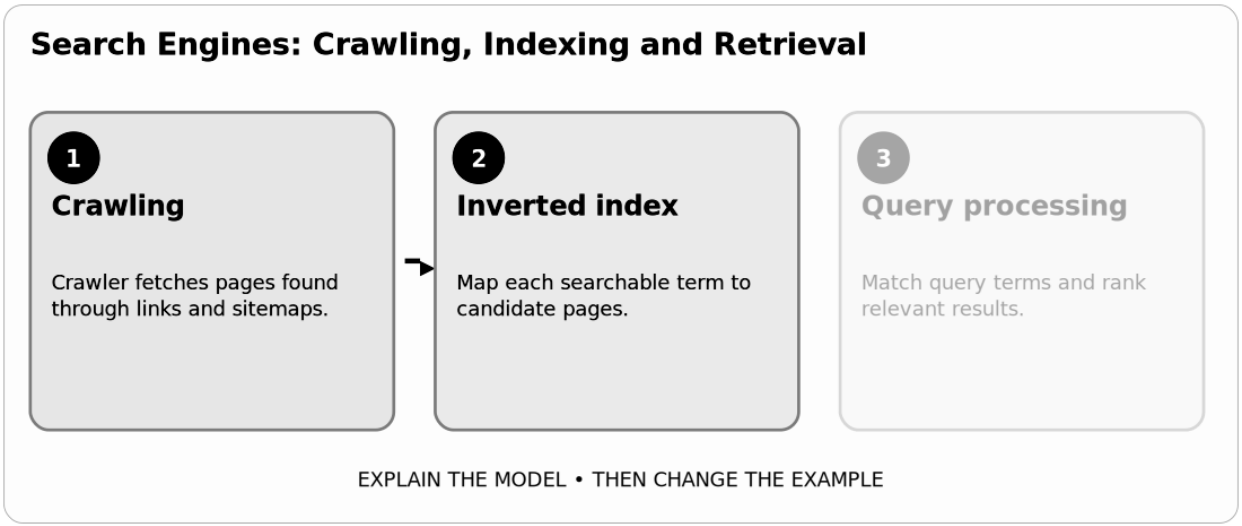
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Search Engines: Crawling, Indexing and Retrieval: Crawling, Inverted index, Query processing

New Learning

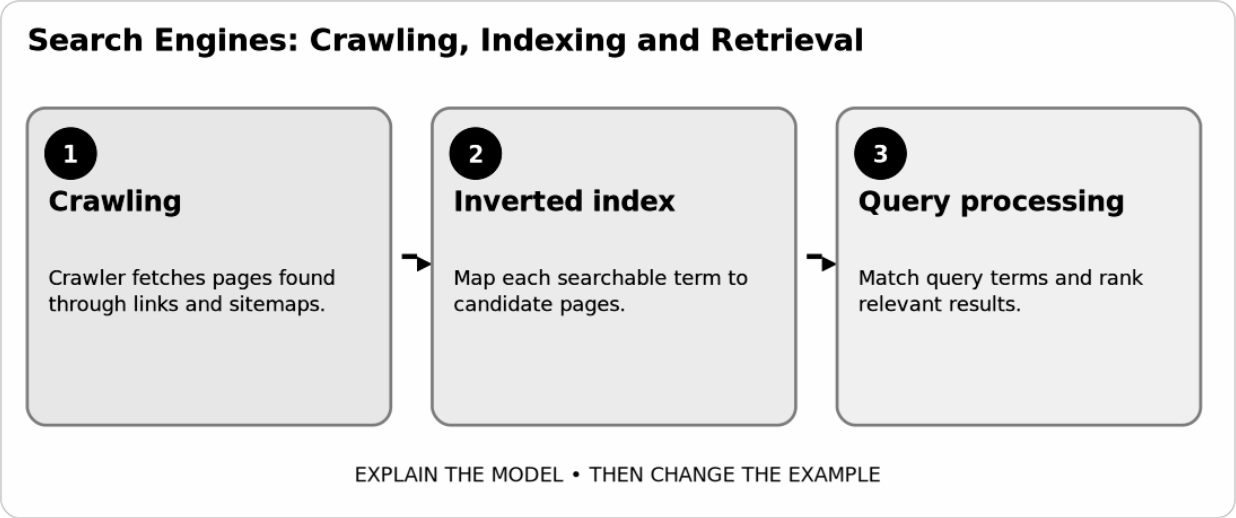


Search Engines: Crawling, Indexing and Retrieval: Crawling, Inverted index, Query processing

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Search Engines: Crawling, Indexing and Retrieval: Crawling, Inverted index, Query processing

I DO — WORKED EXAMPLE

Follow the worked method

Page A has 'binary search' and Page B 'binary tree'. Build a small inverted index.

Tokenise both pages

Identify the input or diagram and predict the first result.

Map terms to page IDs

Follow the second stage and explain its importance.

Match search → A

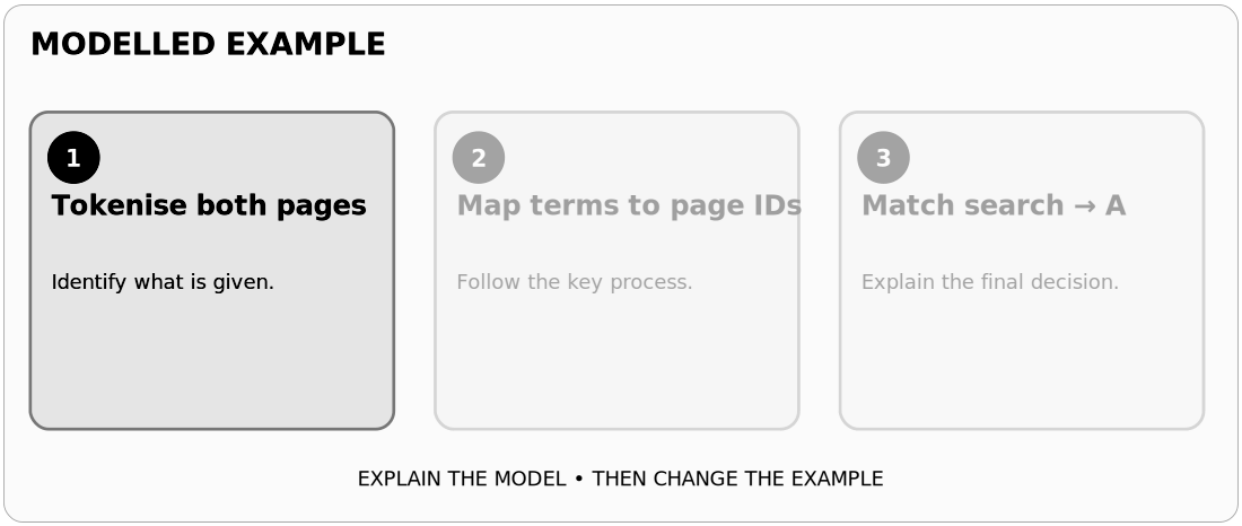
binary→A,B; search→A; tree→B. The query 'binary search' intersects both postings and retrieves A.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Page A has 'binary search' and Page B 'binary tree'. Build a small inverted index.

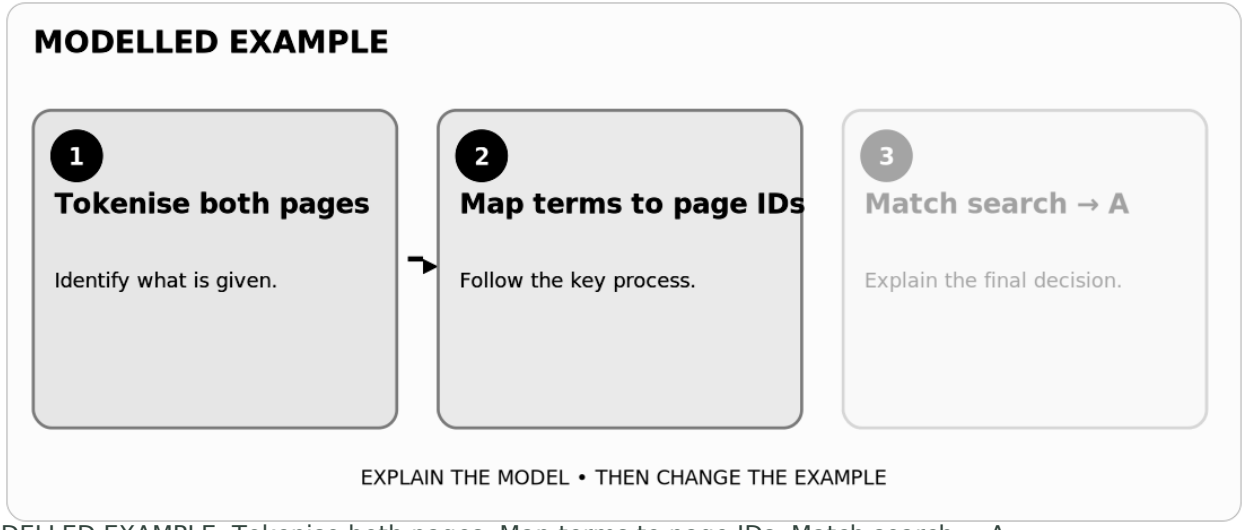
Identify the input or diagram and predict the first result.



MODELLED EXAMPLE: Tokenise both pages, Map terms to page IDs, Match search → A

Page A has 'binary search' and Page B 'binary tree'. Build a small inverted index.

Identify the input or diagram and predict the first result.



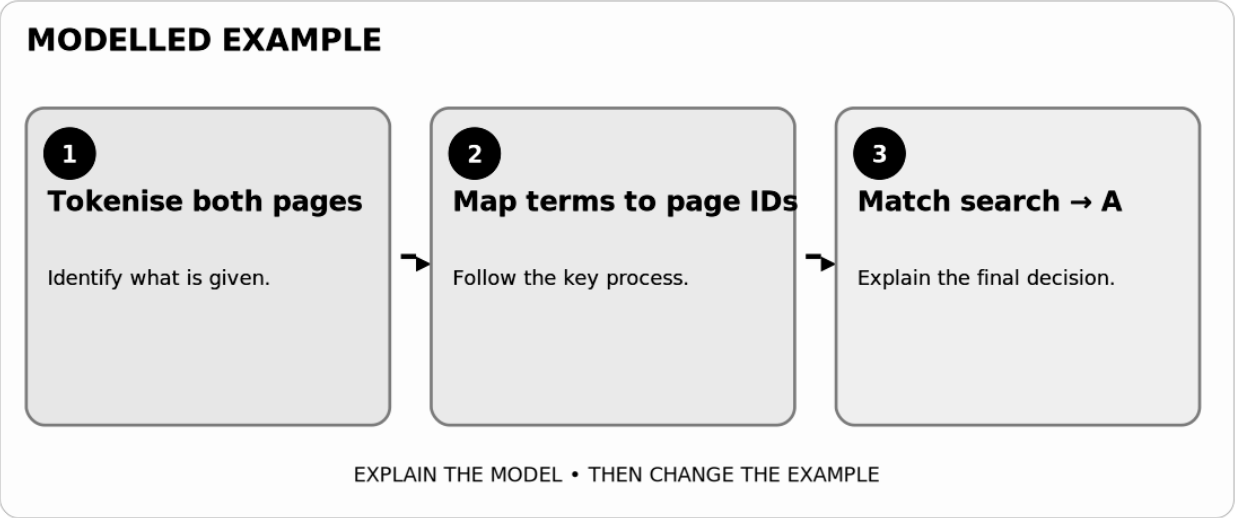
MODELLED EXAMPLE: Tokenise both pages, Map terms to page IDs, Match search → A

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Page A has 'binary search' and Page B 'binary tree'. Build a small inverted index.

Identify the input or diagram and predict the first result.



MODELLED EXAMPLE: Tokenise both pages, Map terms to page IDs, Match search → A

WE DO — GUIDED PRACTICE

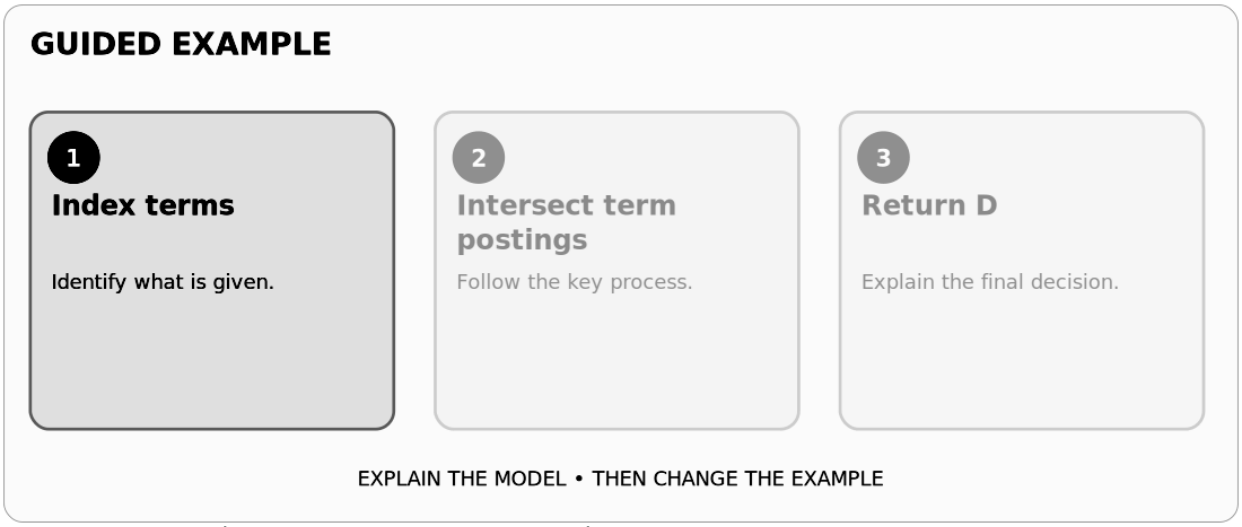
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Page C has 'encryption key' and Page D 'encryption hash'. Which page answers 'encryption hash'?

Identify the input or diagram and predict the first result.



GUIDED EXAMPLE: Index terms, Intersect term postings, Return D

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Index terms Identify the input or diagram and predict the first result.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain crawling and indexing. [6]

Question 2

Index P='cat dog', Q='dog bird' and identify result for 'dog bird'. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Discuss why a new page might not appear in search and why robots.txt is not a security control. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Search indexing and result ranking are the same operation.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

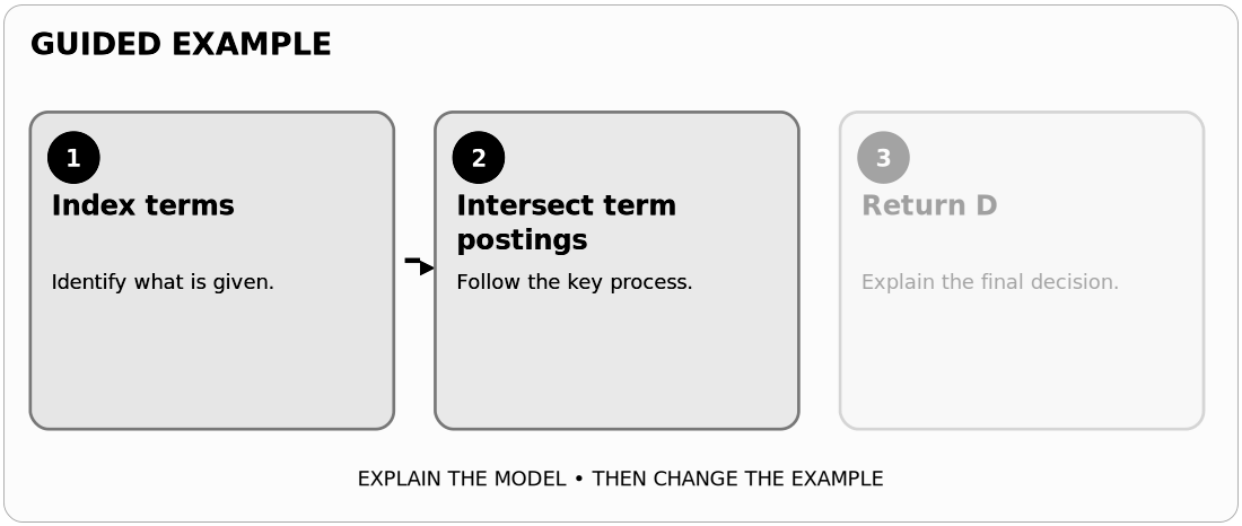
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Page C has 'encryption key' and Page D 'encryption hash'. Which page answers 'encryption hash'?

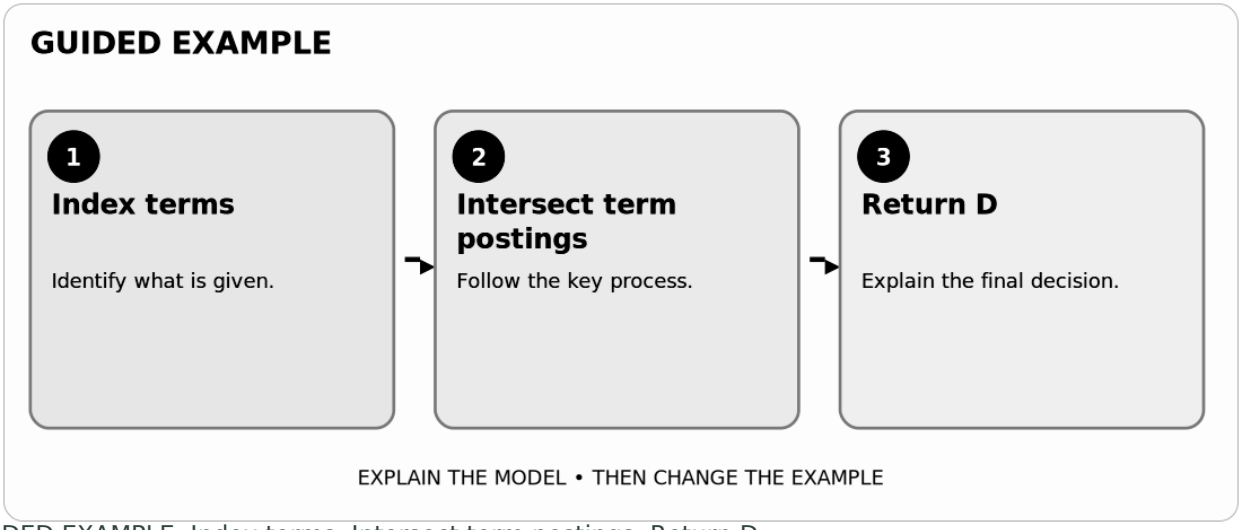
Identify the input or diagram and predict the first result.



GUIDED EXAMPLE: Index terms, Intersect term postings, Return D

Page C has 'encryption key' and Page D 'encryption hash'. Which page answers 'encryption hash'?

Identify the input or diagram and predict the first result.



GUIDED EXAMPLE: Index terms, Intersect term postings, Return D

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Intersect term postings

Follow the second stage and explain its importance.

Return D

Inverted index: encryption→C,D and hash→D. Intersecting returns D. A search engine may rank by further signals after matching.

IMPROVE: Explain one correction to your own reasoning.